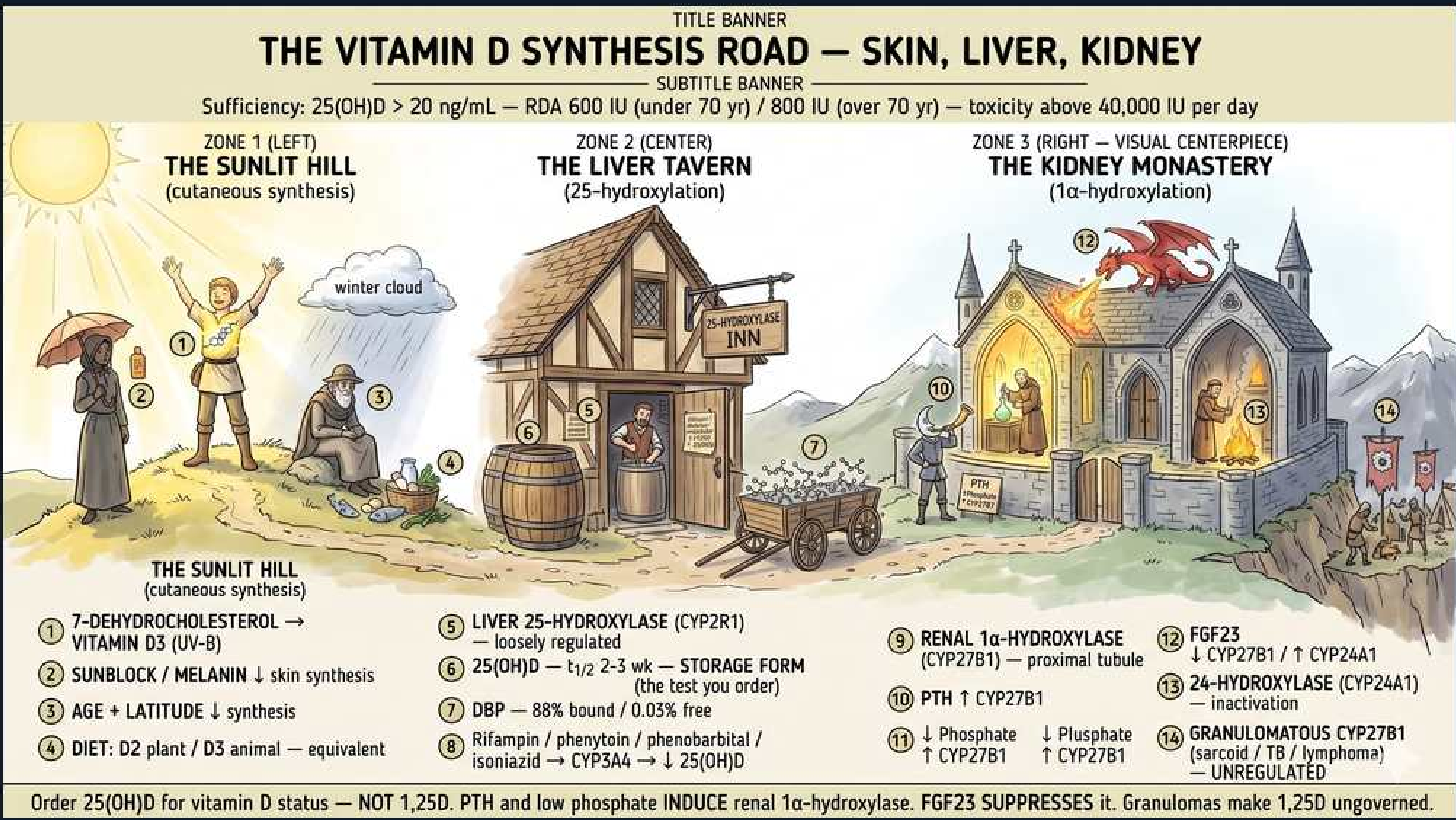
TITLE BANNER
THE VITAMIN D SYNTHESIS ROAD — SKIN, LIVER, KIDNEY
SUBTITLE BANNER
Sufficiency: 25(OH)D > 20 ng/mL — RDA 600 IU (under 70 yr) / 800 IU (over 70 yr) — toxicity above 40,000 IU per day
ZONE 1 (LEFT)
THE SUNLIT HILL
(cutaneous synthesis)
ZONE 2 (CENTER)
THE LIVER TAVERN
(25-hydroxylation)
ZONE 3 (RIGHT — VISUAL CENTERPIECE)
THE KIDNEY MONASTERY
(1α-hydroxylation)
winter cloud
25-HYDROXYLASE INN
PTH
THE SUNLIT HILL
(cutaneous synthesis)
1 7-DEHYDROCHOLESTEROL → VITAMIN D3 (UV-B)
2 SUNBLOCK / MELANIN ↓ skin synthesis
3 AGE + LATITUDE ↓ synthesis
4 DIET: D2 plant / D3 animal — equivalent
5 LIVER 25-HYDROXYLASE (CYP2R1) — loosely regulated
6 25(OH)D — $t_{1/2}$ 2-3 wk — STORAGE FORM (the test you order)
7 DBP — 88% bound / 0.03% free
8 Rifampin / phenytoin / phenobarbital / isoniazid → CYP3A4 → ↓ 25(OH)D
9 RENAL 1α-HYDROXYLASE (CYP27B1) — proximal tubule
10 PTH ↑ CYP27B1
11 ↓ Phosphate ↑ CYP27B1 ↓ Plusphate ↑ CYP27B1
12 FGF23 ↓ CYP27B1 / ↑ CYP24A1
13 24-HYDROXYLASE (CYP24A1) — inactivation
14 GRANULOMATOUS CYP27B1 (sarcoid / TB / lymphoma) — UNREGULATED
Order 25(OH)D for vitamin D status — NOT 1,25D. PTH and low phosphate INDUCE renal 1α-hydroxylase. FGF23 SUPPRESSES it. Granulomas make 1,25D ungoverned.

1. Sufficiency = 25(OH)D >20

WHAT YOU SEE

The subtitle banner reads 'Sufficiency: 25(OH)D >20 ng/mL — RDA 600 IU / 800 IU (over 70) — toxicity above 40,000 IU per day' above the three-stop vitamin D road.

WHAT IT ENCODES

Vitamin D status is judged by serum 25(OH)D (calcidiol), the storage form: >20 ng/mL is generally sufficient for the general population (some societies use >30 for at-risk patients), 12–20 is insufficient, and <12 ng/mL is frank deficiency. RDA is 600 IU/day under age 70 and 800 IU/day over 70; toxicity is uncommon below chronic intakes around 40,000 IU/day. Deficiency causes osteomalacia in adults and rickets in children.

BOARD TRAP

WRONG answer: 'order 1,25-dihydroxyvitamin D to assess deficiency.' The status test is 25(OH)D; 1,25-D has a short half-life and is often NORMAL or even HIGH in deficiency (because secondary hyperPTH drives 1-alpha-hydroxylase). Discriminator: 1,25-D is only ordered for specific puzzles — granulomatous disease, CYP24A1 defects, or hereditary rickets — not routine status.

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2. Skin: 7-DHC -> D3

WHAT YOU SEE

On THE SUNLIT HILL a child throws arms up to a blazing sun (#1), tagged 7-DEHYDROCHOLESTEROL → VITAMIN D3 (UV-B) — sunbathing in the UV rays is the photochemical skin step making D3.

WHAT IT ENCODES

In the skin, UV-B radiation (290–315 nm) converts 7-dehydrocholesterol to pre-vitamin D3, which thermally isomerizes to cholecalciferol (vitamin D3). This is the only step that does NOT require an enzyme — it is photochemical — and it is self-limiting (excess pre-D3 degrades to inert photoproducts, so sun exposure alone cannot cause toxicity). Cutaneous synthesis supplies the majority of vitamin D in most people.

BOARD TRAP

WRONG answer: 'sunbathing can cause vitamin D toxicity.' Excess UV converts pre-vitamin D3 into inactive products (lumisterol/tachysterol), so cutaneous synthesis is self-regulating — toxicity requires oral OVERSUPPLEMENTATION. Discriminator: vitamin D toxicity = ingested megadoses (or granulomatous extrarenal 1,25-D), never sun.

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3. Sunblock / melanin reduce synthesis

WHAT YOU SEE

On the hill, figures shaded by an umbrella, a winter cloud, and an aging man (#2/#3), tagged SUNBLOCK / MELANIN ↓ skin synthesis and AGE + LATITUDE ↓ — everything blocking the sun cuts cutaneous D3.

WHAT IT ENCODES

Anything that blocks UV-B reduces cutaneous D3 synthesis: sunscreen (SPF 30 cuts it ~95%), darker skin (melanin competes for UV-B), high latitude, winter season, indoor lifestyle/institutionalization, and aging skin (less 7-DHC). These factors explain who gets deficiency — the homebound elderly, veiled or dark-skinned individuals at northern latitudes, and exclusively breastfed infants without supplementation.

BOARD TRAP

WRONG answer: attributing an elderly housebound patient's low Ca and bone pain to 'primary hyperparathyroidism.' The setup (reduced sun + poor intake) points to vitamin D deficiency -> osteomalacia with SECONDARY hyperPTH (low/normal Ca, HIGH PTH, low phosphate, high ALP). Discriminator: primary hyperPTH has HIGH calcium; vitamin D deficiency has LOW/normal calcium with high PTH.

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4. Diet: D2 plant / D3 animal

WHAT YOU SEE

A basket of food at the hill base (#4), tagged DIET: D2 plant / D3 animal — equivalent — the food basket showing both dietary vitamin D forms feed into the same road.

WHAT IT ENCODES

Dietary vitamin D comes in two forms: D2 (ergocalciferol) from plants/fungi/fortified foods, and D3 (cholecalciferol) from animal sources (oily fish, egg yolk) and skin synthesis. Both are biologically usable after the same liver and kidney hydroxylations; D3 raises and sustains serum 25(OH)D somewhat more efficiently than D2, which is why D3 is generally preferred for repletion.

BOARD TRAP

WRONG answer: 'D2 (ergocalciferol) is the inactive plant form and cannot raise vitamin D levels.' Both D2 and D3 are pro-hormones that must be hydroxylated twice; both work, though D3 is modestly more potent. Discriminator: the truly 'inactive until hydroxylated' point applies to BOTH — neither is active until the liver and kidney steps occur.

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5. Liver 25-hydroxylase (CYP2R1)

WHAT YOU SEE

THE LIVER TAVERN ('25-HYDROXYLASE INN') with its hanging sign (#5), tagged LIVER 25-HYDROXYLASE (CYP2R1) — loosely regulated — the inn is the liver's first hydroxylation stop.

WHAT IT ENCODES

The LIVER performs the first hydroxylation via 25-hydroxylase (CYP2R1), converting D3/D2 to 25(OH)D (calcidiol). This step is only loosely regulated — it largely tracks substrate availability — which is exactly why 25(OH)D faithfully reflects total body vitamin D stores and is the test you order. Severe liver disease can lower 25(OH)D.

BOARD TRAP

WRONG answer: confusing the liver step with the activating step. The liver makes 25(OH)D (storage, the test); the KIDNEY makes 1,25-D (active, the regulated step). Discriminator: '25 = liver = the test'; '1,25 = kidney = the active hormone.' Don't blame liver disease for failure to ACTIVATE vitamin D — that's a kidney/CKD problem.

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6. 25(OH)D storage form

WHAT YOU SEE

Barrels stacked outside the Liver Tavern (#6), tagged 25(OH)D — t½ 2-3 wk — STORAGE FORM (the test you order) — the storage barrels are the long-lived 25(OH)D you measure.

WHAT IT ENCODES

25(OH)D (calcidiol) is the major circulating/storage form with a long half-life of 2–3 weeks, which is what makes it a stable, reliable index of vitamin D status (unlike 1,25-D, whose half-life is only ~4–6 hours). It circulates bound to vitamin D-binding protein. Because of the long half-life, repletion is monitored after ~3 months.

BOARD TRAP

WRONG answer: '1,25-D, being the active form, is the best marker of stores.' 1,25-D has a SHORT half-life and is tightly regulated, so it is a poor status marker and is often normal/high in deficiency. Discriminator: stable + long half-life + reflects stores = 25(OH)D.

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7. Drugs lower 25(OH)D

WHAT YOU SEE

A figure by the tavern (#8), tagged Rifampin / phenytoin / phenobarbital / isoniazid → CYP3A4 → ↓25(OH)D — the enzyme-inducing drugs that drain the storage barrels.

WHAT IT ENCODES

Hepatic enzyme inducers accelerate vitamin D catabolism via CYP3A4/CYP24-type pathways, lowering 25(OH)D: classic offenders are rifampin, phenytoin, phenobarbital, carbamazepine, and isoniazid. This is why patients on chronic antiepileptics or anti-TB therapy are at risk for drug-induced osteomalacia and need vitamin D supplementation/monitoring.

BOARD TRAP

WRONG answer: 'a seizure patient with bone pain and low 25(OH)D has primary hyperparathyroidism.' Chronic phenytoin/phenobarbital induce vitamin D catabolism -> deficiency/osteomalacia. Discriminator: drug-induced osteomalacia gives LOW Ca/normal-low with HIGH PTH and high ALP — not the HIGH calcium of primary hyperPTH.

8. Kidney 1-alpha-hydroxylase (CYP27B1)

WHAT YOU SEE

THE KIDNEY MONASTERY church on the right (#9), tagged RENAL 1α-HYDROXYLASE (CYP27B1) — proximal tubule — the monastery is the kidney's tightly-regulated activation step making 1,25-D.

WHAT IT ENCODES

The KIDNEY proximal tubule performs the second, ACTIVATING hydroxylation via 1-alpha-hydroxylase (CYP27B1), converting 25(OH)D to 1,25(OH)2D (calcitriol) — the rate-limiting, tightly regulated step and the active hormone. It is induced by PTH and hypophosphatemia and suppressed by FGF23 and high Ca/PO4. In CKD this step fails, causing low 1,25-D, hypocalcemia, and secondary hyperparathyroidism (renal osteodystrophy).

BOARD TRAP

WRONG answer: 'CKD patients need plain cholecalciferol to fix their low calcium.' Plain D can't be activated without functional renal 1-alpha-hydroxylase — they need ACTIVE vitamin D (calcitriol/paricalcitol). Discriminator: the kidney = activation; renal failure = give the active form, not the precursor. Also note loss-of-function CYP27B1 = vitamin D-dependent rickets type 1 (low 1,25-D despite adequate 25-D).

9. PTH + low PO4 turn it ON

WHAT YOU SEE

A monk feeding the monastery's fire (#10/#11), tagged ↓Phosphate / PTH ↑ CYP27B1 — PTH and low phosphate stoke the activating-enzyme fire ON.

WHAT IT ENCODES

Two signals INDUCE renal 1-alpha-hydroxylase to make more active 1,25-D: rising PTH and falling serum phosphate (hypophosphatemia). This closes the loop — PTH not only mobilizes Ca from bone and reabsorbs it in the distal tubule but also boosts 1,25-D to raise intestinal Ca absorption. Conversely, high Ca, high phosphate, and FGF23 turn the enzyme OFF.

BOARD TRAP

WRONG answer: 'PTH suppresses vitamin D activation.' PTH STIMULATES 1-alpha-hydroxylase (turns it on); FGF23 is the suppressor. Discriminator: ON switches = PTH + low phosphate; OFF switches = FGF23 + high Ca/PO4 + 1,25-D itself (negative feedback).

10. FGF23 suppresses 1,25-D

WHAT YOU SEE

A red DRAGON breathing fire over the monastery (#12), tagged FGF23 → ↑CYP24A1 / suppresses CYP27B1 — the dragon is FGF23 shutting off vitamin D activation and wasting phosphate.

WHAT IT ENCODES

FGF23 (a phosphatonin from osteocytes, acting via Klotho co-receptor) is the body's anti-phosphate, anti-vitamin-D hormone: it makes the kidney WASTE phosphate (down-regulates NaPi cotransporters), SUPPRESSES 1-alpha-hydroxylase (CYP27B1), and INDUCES 24-hydroxylase (CYP24A1) — lowering 1,25-D. It is central to CKD-MBD (rises early in CKD) and to phosphate-wasting rickets (X-linked hypophosphatemic rickets from PHEX mutation, and tumor-induced osteomalacia).

BOARD TRAP

WRONG answer: 'tumor-induced osteomalacia is caused by hypercalcemia and high vitamin D.' It is driven by tumoral FGF23 overproduction -> renal phosphate wasting, LOW phosphate, and LOW/inappropriately-normal 1,25-D. Discriminator: FGF23 excess = low phosphate + low 1,25-D (look for a small mesenchymal tumor); PTH excess = high Ca + low phosphate but HIGH 1,25-D.

11. 24-hydroxylase inactivation

WHAT YOU SEE

A monk dousing/extinguishing the fire (#13), tagged 24-HYDROXYLASE (CYP24A1) — inactivation — the OFF-switch monk who degrades active vitamin D.

WHAT IT ENCODES

24-hydroxylase (CYP24A1) is the OFF switch that degrades both 25-D and 1,25-D into inactive water-soluble metabolites (e.g., calcitroic acid). It is induced by 1,25-D itself (feedback) and by FGF23, protecting against vitamin D excess. This is the catabolic counterweight to 1-alpha-hydroxylase.

BOARD TRAP

WRONG answer: 'an infant with hypercalcemia, low PTH, and suppressed 24-hydroxylase activity just has too much dietary vitamin D.' Loss-of-function CYP24A1 mutations cause idiopathic INFANTILE HYPERCALCEMIA — the inactivating enzyme is broken, so vitamin D cannot be cleared. Discriminator: hypercalcemia + low PTH + low 24,25-D / high 25-D:24,25-D ratio = CYP24A1 defect.

12. Granulomatous extra 1,25-D

WHAT YOU SEE

A figure with a banner outside the monastery walls (#14), tagged GRANULOMATOUS CYP27B1 (sarcoid / TB / lymphoma) — UNREGULATED — rogue extrarenal activation that ignores normal feedback.

WHAT IT ENCODES

Activated macrophages within granulomas express their own CYP27B1 (1-alpha-hydroxylase) that is NOT subject to normal feedback (no suppression by high Ca or low PTH), so they overproduce 1,25-D. Seen in sarcoidosis, tuberculosis, other granulomatous infections, and some lymphomas, this causes hypercalcemia/hypercalciuria with a SUPPRESSED PTH and elevated 1,25-D. It is glucocorticoid-responsive (steroids shut off macrophage 1-alpha-hydroxylase).

BOARD TRAP

WRONG answer: treating sarcoid hypercalcemia with vitamin D, or attributing it to primary hyperPTH. Granulomatous hypercalcemia has LOW PTH with HIGH 1,25-D and responds to GLUCOCORTICOIDS (also avoid sun/vitamin D). Discriminator: high 1,25-D + low PTH + suppressed = extrarenal granulomatous production; high PTH = primary hyperPTH.

13. Order 25(OH)D not 1,25-D

WHAT YOU SEE

The bottom banner reads 'Order 25(OH)D for vitamin D status — NOT 1,25-D. PTH and low phosphate INDUCE renal 1α-hydroxylase. FGF23 SUPPRESSES it. Granulomas make 1,25-D ungoverned.'

WHAT IT ENCODES

Summary of the road: STATUS is measured by 25(OH)D (liver product, long half-life, reflects stores); ACTIVITY is set by the kidney 1-alpha-hydroxylase, turned ON by PTH and low phosphate and OFF by FGF23 and 24-hydroxylase. The active hormone 1,25-D drives intestinal Ca/PO4 absorption. Order 1,25-D only for specific scenarios (granulomatous disease, hereditary/phosphate-wasting rickets, CYP24A1 defects).

BOARD TRAP

WRONG answer: 'use 1,25-D to screen for deficiency.' This is the single most-tested vitamin D error — 1,25-D is short-lived, tightly regulated, and frequently normal/high in true deficiency. Discriminator: deficiency workup = 25(OH)D; the regulated active form is only diagnostically useful in the specific syndromes above.